



International Innovation Challenge

2.0

September 08-09, 2025

Release of problem statements

Venue: Smt. Vasanti R Pai Auditorium

Date: September 08, 2025

1. Image based breed recognition for cattle and buffaloes of India
2. AI based development of Laser based QR Code marking on 'track fittings on Indian Railways'.
3. Development of Sensor for Detection Of Microplastics
4. Automated Compliance Checker for Legal Metrology Declarations on E-Commerce Platforms
5. Smart Tourist Safety Monitoring & Incident Response System using AI, Geo-Fencing, and Blockchain-based Digital ID
6. Develop a blockchain-based system for botanical traceability of Ayurvedic herbs, including geo-tagging from the point of collection (farmers/wild collectors) to the final Ayurvedic formulation label.
7. Develop API code to integrate NAMASTE and or the International Classification of Diseases (ICD-11) via the Traditional Medicine Module 2 (TM2) into existing EMR systems that comply with Electronic Health Record (EHR) Standards for India.
8. Intelligent Pesticide Sprinkling System Determined by the Infection Level of a Plant
9. Low-Cost smart transportation solution for Agri produce from remote farms to nearest motorable road in NER Region
10. Low-Cost smart transportation solution for Agri produce from remote farms to nearest motorable road in NER Region
11. Gamified Environmental Education Platform for Schools and Colleges
12. Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases in Rural Northeast India
13. Digital Platform for Centralized Alumni Data Management and Engagement
14. AI-Based Internship Recommendation Engine for PM Internship Scheme
15. Smart Waste Segregation and Recycling System
16. Improved Onion storage technology for enhancing shelf life of onions
17. Designing and development of an application for on spot assessment of Roof Top Rain water harvesting and artificial recharge potential and size of the RTRWH and AR.
18. AI-Based Farmer Query Support and Advisory System
19. To design and develop an indigenous, secure, and ruggedized Customer Premises Equipment (CPE) system to enable encrypted and reliable internet connectivity onboard

Indian naval ships under the Secure Internet for onboard Naval Ships (SIFON) framework in coordination with TSP.

20. Design AI-powered bots for small farmers to aid with soil testing, pest detection, crop management, irrigation, and accessing financial aids.
21. Develop solutions for tracking immunization services, monitoring temperature-sensitive goods for vaccines and medicines during transport, and creating apps for disease outbreak management.
22. Develop a mobile application for tracking district officials' field visits and inspections using geo-tagging and digital authentication.
23. Develop innovative assistive technologies and inclusive infrastructure to empower visually impaired individuals with enhanced navigation, mobility, and independence, thereby improving their social interaction and overall quality of life.
24. Develop a solution for connecting farmers with local markets, enabling real-time price discovery, efficient supply chain management, and fair pricing for agricultural produce.
25. Create a user-friendly software solution to map government schemes to beneficiaries based on their socio-economic background? Ensure real-time access and a citizen-centric design for efficient grievance redressal.
26. develop a unique and efficient mechanism to automatically stop the transfer of social security funds to deceased under various schemes? The solution should ensure timely identification of deceased beneficiaries, leveraging integration with death registration databases, Aadhaar authentication, or periodic life certification through digital methods to prevent misuse and save government funds.
27. Create accessible VR and AR-based tools to enhance hands-on, industry-relevant training for students.
28. Develop an integrated "All in one" mobile app for MUJ including smart parking system with pre-notification and navigation to vacant parking space, smart and easy navigation within MUJ campus, checking availability of MUJ employees with location and phone number, alumni connection, complaints and tracking for general services.
29. AI-Driven Public Health Chatbot for Disease Awareness

Description

Create a multilingual AI chatbot to educate rural and semi-urban populations about preventive healthcare, disease symptoms, and vaccination schedules. The chatbot should integrate with government health databases and provide real-time alerts for outbreaks.

Expected Outcome

A chatbot accessible via WhatsApp or SMS, reaching 80% accuracy in answering health queries and increasing awareness by 20% in target communities.

Technical Feasibility

Built using NLP frameworks (e.g., Rasa, Dialogflow) with APIs for health data integration, deployable on cloud platforms for scalability.

30. Quality Control for Medical Imaging Studies

Description

Build an AI-powered pipeline that automatically checks CT scans for resolution, completeness, and anonymization compliance before they are used for training AI models.

Expected Outcome

A quality-check dashboard flagging failed studies with reasons (low resolution, missing slices, PHI leakage) and achieving >90% accuracy in anomaly detection.

Technical Feasibility

Leverage PyTorch/TensorFlow models for resolution detection, regex + NLP for report validation, integrated into a Databricks/Azure workflow.

31. Multi-Label Chest X-Ray Findings Detection

Description

Most X-ray images contain multiple findings (e.g., pneumonia + effusion). Create an AI system that can detect multiple abnormalities simultaneously rather than just normal/abnormal.

Expected Outcome

A model that outputs probabilities for at least 3–5 disease categories per X-ray, with a dashboard to visualize predictions and confidence scores.

Technical Feasibility

Public datasets (NIH ChestX-ray14, CheXpert) can be used. Pretrained models (ResNet/DenseNet) can be fine-tuned within 36 hours. Grad-CAM visualization can be added for interpretability.

32. Real-Time Imaging Workflow Orchestration

Description

In hospitals, multiple scans arrive every minute. Design a workflow tool that ingests DICOM files, routes them through anomaly detection models, and pushes prioritized cases to a mock “radiologist dashboard.”

Expected Outcome

An end-to-end pipeline simulation: input (DICOM/PNG) → AI model → results sorted by urgency. A simple web dashboard shows the case queue with timestamps.

Technical Feasibility

Can be simulated using Flask/Django + pre-trained models. Use open-source DICOM libraries (pydicom). Teams focus on workflow automation + AI integration.

33. Privacy-Preserving Medical Image Sharing

Description

Medical images often contain sensitive information. Build a pipeline that automatically removes text labels and anonymizes metadata while preserving diagnostic quality.

Expected Outcome

A prototype that takes in medical images with patient names/IDs and outputs anonymized versions, along with a report comparing original vs anonymized images.

Technical Feasibility

Use OCR (Tesseract/EasyOCR) for text detection and image inpainting. Remove DICOM metadata fields using pydicom.

34. AI-Powered Multi-Modal Classroom Engagement Analyzer (Non- Healthcare)

Description

Build an AI tool that analyzes classroom videos and determines if students are engaged, distracted, or absent using both facial expressions (vision) and voice/text cues (audio/NLP).

Expected Outcome

A demo that labels engagement levels of students in a short clip and a dashboard with engagement statistics over time.

Technical Feasibility

Open datasets: AffectNet (facial expressions), RAVDESS (speech emotion). Use pretrained emotion detection + late-stage fusion for multi-modal analysis.

35. AI-Powered Spine Posture s Curve Estimation from Human Body

Description

Create a computer-vision system that estimates spine alignment and curvature from normal RGB/depth images (webcam or phone camera). The system should identify body keypoints (shoulders, hips, back) and approximate spinal curves to detect abnormal postures, scoliosis tendencies, or long-term musculoskeletal risks.

Expected Outcome

A real-time posture monitoring tool that achieves >85% accuracy in estimating spinal alignment compared to physiotherapist assessments, and provides visual feedback for correction (e.g., “leaning left”, “slouched posture”).

Technical Feasibility

Use pose-estimation frameworks (MediaPipe Pose, OpenPose, or SMPL model fitting) to extract 2D/3D skeletal landmarks. Apply curve-fitting (splines/polynomials) to approximate the spine’s natural curve and compute posture deviation metrics.

Deployable as a desktop/mobile app for physiotherapy, ergonomics, or fitness use cases.

36. Dashboard development using AI-ML techniques to interpolate tropospheric Precipitable Water (PW) content using zenith-wet delay from GNSS observations

Background:

Troposphere is a critical layer of earth’s atmosphere which influences weather patterns. Troposphere holds Precipitable Water (PW) and affects the transmission of GNSS signals. GNSS reference stations capture signals from multiple satellites and have observations, which can help estimate tropospheric wet content. If the precise coordinates of reference station and satellites are known, the GNSS reference station can compute zenith-wet delay. The zenith-wet delay along line-of-transmission of GNSS signal can help understand the distribution of PW in a given area and help forecast rainfall weather alerts.

Detailed Description:

In this task, the zenith-wet delay, observed over 24 hours to 7 days, at multiple GNSS stations shall be provided. Additionally, the elevation and azimuth of GNSS satellites shall be provided. The challenge is to develop an AI/ML-powered algorithm to interpolate tropospheric wet delay over a given region of GNSS network. The software solution shall

- Accept zenith-wet delay observations from multiple stations
- Compute temporal wet delay between multiple stations based on elevation and azimuth of GNSS satellites
- Apply AI-ML techniques to improve interpolation accuracy

- Display results on a dashboard

Expected Solution:

The developed solution shall provide:

- A dashboard which shall show the interpolated wet delay at any point in the given area
- Accept third-party observations of wet-delay and compute error in the interpolation results
- Extend the algorithm to extrapolate wet delay in regions beyond the given GNSS network

37. Ultra-Extreme Image Compression for Low-Bandwidth Transmission Using AI-Based Super-Resolution (SR)

Background

Super-resolution is a technique in computer vision that enhances the resolution and quality of low-resolution images or videos using deep learning. It reconstructs fine details, sharpens edges, and improves clarity. This technology is crucial for applications like medical imaging, satellite imagery, surveillance, and digital restoration where detail preservation is critical.

Detailed Description:

Develop an AI-powered image compression and reconstruction system that does following:

- Input: Takes a high-resolution image (~1 MB).
- Compression: Reduces it to a maximum of 1 KB for transmission or storage.
- Reconstruction: Recovers a high-resolution image (~1 MB) from the 1 KB file with minimal perceptual distortion.
- Output Quality: The reconstructed image should be visually close to the original, preserving key structures, colours, and features, while keeping Mean Opinion Score (MOS) and perceptual similarity metrics (LPIPS, MS-SSIM) within acceptable thresholds.

Expected Solution:

- Encoder: Converts input image into a ≤ 1 KB compressed representation.
- Decoder: AI model that reconstructs the image at full resolution.
- Performance Report:
 - Compression ratio achieved.
 - MS-SSIM, LPIPS, and MOS scores vs. baselines (JPEG, AVIF at same size).
- Demonstration Application: Web or desktop app showing side-by-side comparison of original vs. reconstructed images. Any 1MB image can be taken

38. Cross Modal SAR-Optical Image Matching/Registration

Background

SAR and optical imagery provide complementary information about the Earth's surface. SAR is sensitive to surface roughness and dielectric properties, while optical imagery provides information about surface reflectance and colour. Combining these modalities offers improved capabilities for various applications including land cover classification, change detection, disaster monitoring, and urban planning. However, directly matching (Image Registration) these images is challenging due to differing sensor characteristics, geometric distortions, and viewing geometries.

Detailed Description

The solution should explore and implement advanced feature extraction and matching techniques to overcome these challenges and establish a strong correspondence between SAR and Optical imagery. The proposed solution should implement following aspects:

- Data Acquisition & Preprocessing: Download and preprocessing of relevant SAR and optical image pairs from the Cloudsen12 dataset, including geometric correction, co-registration, and radiometric calibration.
- Feature Extraction: Investigate and implement robust feature extraction techniques suitable for both SAR and Optical images. This will include exploring both conventional (e.g., SIFT, HOG, corner detectors) and deep learning-based feature descriptors (e.g., SiLK, SuperPoint, LoFTR, ViT).
- Geometric Transformation Estimation: Estimate the geometric transformation (e.g., affine, projective) between the matched features to align the SAR and optical images.
- Performance Evaluation: Quantitatively evaluate the performance of the developed algorithm using standard metrics such as:
 - *Matching Accuracy*: Precision, Recall, F1-Score of matched features.
 - *Geometric Accuracy*: Root Mean Square Error (RMSE) of control points.
- Comparative Analysis: Compare the performance of different developed conventional & ML/DL based feature extraction and matching techniques.

Expected Solution

Development of a robust and accurate algorithm for matching Synthetic Aperture Radar (SAR) and optical images, leveraging the rich and readily available Cloudsen12 dataset.

Dataset Link: <https://cloudsen12.github.io/>

39. AI/ML Optical Flow based Frame interpolation of Satellite Images for generation of High Temporal Resolution data

Background:

Satellite images are often captured at fixed intervals (e.g., every 30 minutes for geostationary satellites like INSAT, every 10 minutes for Himawari/GOES satellites or every few days for polar-orbiting satellites). This leads to limitation in temporal resolution that restricts near real-time monitoring of dynamic phenomena such as cyclones, thunderstorms, floods, and rapid land changes. Traditional temporal interpolation is limited with incorrect results (blurred images & other artefacts) and they fail to capture fast-moving or non-linear changes in cloud dynamics.

This problem proposes to develop an AI/ML based Optical Flow frame interpolation technique to generate intermediate frames between two consecutive satellite images to effectively enhance temporal resolution, enabling more frequent observations without additional satellite resources.

Detailed Description:

The developed software should include and accomplish the following steps:

- Develop an optical flow-based model to estimate pixel-wise motion vectors between consecutive satellite frames.
- Generate synthetic intermediate frames using AI-driven frame interpolation methods (e.g., Deep Optical Flow, RAFT, Super SloMo).
- Improve temporal resolution of satellite imagery (e.g., from 30 minutes to 15 minutes and further to 7.5 minutes interval or equivalent).
- Validate results against real higher temporal resolution datasets (like Himawari/GOES-R) using image similarity metrics such as SSIM, MSE, PSNR, FSIM, etc.

Expected Solution:

Solution for the challenge should include a prototype for AI-powered frame interpolation for satellite imagery that should comprise of:

1. Frame Interpolation: Takes two consecutive satellite images and generates intermediate frames. Should include:
 - a. Data collection of satellite datasets from online (INSAT-3D/3DR, Himawari-8, GOES-R)
 - b. Optical Flow Estimation – can explore and use AI/ML models
 - c. Frame Interpolation – can explore and use interpolation networks for synthesis of new frames
2. Visualization: Develop a dashboard showing original vs. interpolated satellite animations. Should include:
 - a. After interpolation of a set of satellite frames, the output should be displayed with time-lapse animations.
 - b. Animations to be present both for ground truth (original satellite frames) and also for interpolated frames.

3. Evaluation: Generate a report comparing results with ground truth (existing frames). Should include:
 - a. Comparison of generated frames with the ground truth using metrics of SSIM, MSE, FSIM, etc. Include plots of the results
 - b. Any other suitable metrics can also be explored and used. Metrics should be suitable to capture the cloud movements.

Hence, the developed solution should have the capability of:

1. Improving temporal density of satellite data without new hardware.
2. A deployable pipeline (Python-based, using TensorFlow/PyTorch/any other framework + OpenCV).

40. Vision based Intelligent RPA agent

Background:

Most RPA systems today work by using fixed selectors (like XPath, control IDs, or DOM paths). These tend to break the moment the interface changes even slightly — which means a lot of manual re-configuration. Now with the developments done in the field of vision transformer models and a number of powerful open-source vision-language models, it has become feasible with reasonable resources to “look at” a screen and understand what’s on it. This has led to development of visual RPA agents that can interact with digital interfaces — whether it’s a website, desktop screen or a dashboard — based purely on what they see and a natural language instruction.

Detailed Description:

In this challenge, students need to build a generic visual RPA agent that interacts with digital interfaces using vision and natural language. Instead of relying on predefined DOM rules or CSS selectors, the agent should:

- Take a screenshot of the current interface (web/desktop)
- Use an existing Vision Transformer model (e.g., DETR, DINO, Swin, SAM, etc.) to detect actionable UI elements on the screen — like buttons, input fields, dropdowns, links, etc. The existing models may be further fine-tuned for improved accuracy.
- Run an open-source OCR tool to extract the text from those elements.
- Read a natural language instruction from the user and match that instruction to the correct visual element using a multi-modal model (e.g. CLIP, sentence transformers, etc.).
- Perform the action using only open-source automation tools (e.g., Selenium, Playwright, PyAutoGUI etc.).

RPA agent should work on Web applications and at least one of the operating systems (Windows or Linux)

Only self-developed or open-source software should be used for the development. Even if APIs to language models are used, the model behind that API should be open-source and available for offline deployment.

Expected Solution:

The final submission should include:

- A perception module that identifies actionable UI elements using a ViT-based model
- An OCR component to read any visible text
- A matching module to link a natural language instruction with the correct on-screen element
- An automation layer that performs the action on the real interface using open-source tools
- A short demo showing the agent successfully completing a multi-step task (e.g., filling and submitting a form, navigating a dashboard, downloading a report, etc.) based only on natural language input.

41. Post Quantum Cryptography (PQC) based TLS Termination Reverse Proxy**Background**

Transport Layer Security (TLS) is the backbone of secure communication on the internet. Current TLS 1.3 connections rely on classical cryptography (RSA, ECDSA, X25519), which are vulnerable to future quantum computers. To prepare, the cryptographic community has standardized post-quantum algorithms like Kyber (KEM) and Dilithium (signatures). Libraries such as OpenSSL 3 with oqs-provider and liboqs already support PQC-enabled TLS handshakes. The challenge is to apply these libraries to create a quantum-safe TLS termination proxy that can be implemented in real-world environments.

Detailed Description

Your task is to implement a TLS termination proxy that uses post-quantum cryptography for secure key exchange and authentication. The proxy should accept incoming TLS 1.3 connections secured with PQC (Kyber, Dilithium, or hybrid classical+PQC), then forward the decrypted traffic to the backend service (For demo purpose local service can be used).

- Configure the TLS handshake to use Kyber KEM for key exchange and optionally Dilithium for certificates/signatures.
- Provide a hybrid mode (classical + PQC combined) to maintain backward compatibility.
- Ensure interoperability with standard clients (e.g., curl, browser) when using a custom CA certificate.
- Log and compare latency and throughput benchmarks for 100 handshakes under classical TLS vs PQC TLS.
- Supply a fallback configuration that allows classical-only TLS

Expected Solution

A complete submission will include:

- Working Proxy: Accepts TLS 1.3 PQC-enabled connections and forwards plaintext traffic to a backend service.
- Configuration & Scripts: Easy setup with Dockerfile or one-line run command.
- Documentation: README explaining architecture, libraries used, PQC algorithms enabled, and setup steps for testing.
- Tests & Benchmarks: Demonstrate successful PQC handshakes, failed handshake cases (invalid certs), and performance metrics vs classical TLS.
- Security Notes: Explanation of hybrid mode, key/certificate handling, and potential deployment considerations.

42. AI-Powered Offline Organizational Chatbot

Background:

Organisations generate and store vast amounts of Employees information across Administration/ HR systems and employee portals. Traditional systems require manual navigation through multiple applications. A local AI-driven chatbot can simplify this by allowing employees, managers and HR to query data conversationally. Further, deploying such systems requires ensuring **data security**, **role-based access** and the solution must run offline without dependency on cloud services.

Description:

Design and develop an offline chatbot that can answer employee queries by retrieving information from organizational databases. The chatbot must:

- Convert organizational data into a vector database to enable efficient and accurate question-answering capabilities.
- Implement role-based access control (RBAC) with different roles such as employees, manager, administration etc
- Accuracy: The chatbot should demonstrate high factual accuracy and relevance in its responses.

Expected Solution:

Participants are expected to deliver a fully functional offline chatbot with the following components:

- **Web Chatbot Interface**

- **Data Ingestion and Vectorization:** A mechanism to convert JSON data into a vector database suitable for semantic search and question answering.
- **Offline LLM Integration:** A locally hosted LLM capable of processing natural language queries and generating accurate responses based on the vector database.
- **Role-Based Access Control:** A secure system to authenticate users and restrict data access based on their role (Employee, Manager or Administration/HR).
- **Query Processing:** A pipeline to process natural language queries, retrieve relevant data from the vector database and generate human-readable responses.
- **On-Premises Deployment:** The entire solution, including the LLM and vector database, must run locally on a single machine or a small cluster without internet dependency.

43. Next-Gen Desktop Based Secure Data Vault

Background:

Organizations manage large volume of sensitive documents, reports, and research data. Protecting this information with high security and confidentiality is a critical requirement. At the same time, the organization needs to share specific data among teams in a controlled manner. Existing cloud or browser-based vaults raise concerns about data privacy, external dependencies, and compliance. Therefore, the organisation requires a desktop-based secure vault solution that can run reliably within its infrastructure. Such a system would ensure confidentiality, controlled sharing, and compliance while empowering employees with secure, easy access to critical data.

Description:

The proposed system should function as a desktop application capable of storing all documents in encrypted form on the local machine while also maintaining a secure copy on the server. Any changes either made locally or on the server must be synchronized in both directions, ensuring data consistency across environments. Since the vault will operate within the organization intranet, it should support secure communication channels with the server while remaining fully functional in offline mode, synchronizing data automatically when reconnected.

To safeguard sensitive information, the system must use standard encryption algorithms and incorporate measures to defend against ransomware attacks, such as secure backups or version-controlled storage. It should also provide role-based and group-based access controls, allowing documents to be shared with authorized individuals or teams. Whenever documents are shared or updated, the concerned members should receive timely notifications or email alerts. Additionally, the solution must support audit trails and logging, ensuring every action—such as access, modification, or sharing—can be traced for accountability.

The ultimate goal is to deliver a secure, resilient, and easy-to-use desktop vault that protects sensitive organizational data while enabling safe collaboration within teams.

Expected Solution:

Participants are expected to design and prototype a **desktop-based secure data vault** that demonstrates the following capabilities:

- **Cross-Platform Support** – The application should run seamlessly on **any major operating system** (Windows, Linux, macOS). Portability and compatibility are essential for organization-wide adoption.
- **Secure Local & Server Storage** – All data/documents must be encrypted before storage, both on the local machine and the server. The solution should ensure that even if files are accessed outside the vault, they remain unreadable without proper authorization.
- **Two-Way Synchronization** – Any modification made on the local machine or server should automatically sync in both directions when connected to the intranet, while ensuring conflict resolution and version control.
- **Offline Functionality** – The vault should remain fully operational even without intranet connectivity. Users should be able to view, edit, and save documents locally, with updates syncing once connectivity is restored.
- **Strong Encryption & Security** – Use of standard, proven encryption algorithms (e.g., AES-256, RSA) to secure documents. The system should also demonstrate basic protection measures against ransomware-style attacks, such as immutable backups or version history.
- **Role-Based & Group-Based Access** – The vault should allow administrators to create roles and groups (e.g., directors, team heads, staff) and grant permissions accordingly. Documents can be shared securely with groups or individuals.
- **Notifications & Alerts** – Whenever documents are shared, updated, or accessed, the concerned members should be notified through in-app notifications or email alerts.
- **Audit & Logs** – The solution must provide logs for access, modifications, and sharing, ensuring accountability and traceability.
- **Ease of Deployment** – The desktop application should be portable, lightweight, and easy to deploy within an organization's intranet without requiring complex setup.

Participants are free to choose the technology stack, but the final solution must demonstrate **security, usability, resilience, and cross-platform compatibility** in handling sensitive organizational documents

44. AI/ML-Driven Intelligent Software Code Analysis for Enhanced Quality and Security

Background:

Software code analysis is crucial for detecting bugs, vulnerabilities, and inefficiencies early in the development lifecycle. Traditional static and dynamic analysis tools often struggle with complex codebases, evolving project contexts, and scale. AI/ML techniques promise to improve automated code analysis by understanding code semantics, predicting defects, and optimizing code quality. However, challenges

remain, such as handling large project contexts, adapting to diverse coding styles, ensuring accuracy without overwhelming developers with false positives, and maintaining explainability and trust in AI suggestions. Especially in dynamic and resource-constrained environments, an intelligent AI/ML solution tailored to practical developer needs can significantly improve software quality assurance.

Detailed Description:

The challenge is to develop an AI/ML-powered software code analysis platform that can:

- Analyze large and complex codebases by breaking them into manageable units while preserving context.
- Detect bugs, security vulnerabilities, code smells, and performance bottlenecks using intelligent models trained on diverse coding patterns.
- Adapt to project-specific coding standards and styles automatically, providing consistent and relevant analysis results.
- Offer explainable insights and recommendations to help developers understand and trust the AI-based findings.
- Integrate seamlessly with existing development environments and supporting multiple programming languages and frameworks.

Expected Solution:

An AI/ML-based smart code analysis tool that leverages advanced machine learning techniques and natural language processing to provide automated, adaptive, and explainable code reviews. It should include features like:

- Modular analysis for scaling to large projects.
 - Real-time detection and prioritization of critical code issues.
 - Customization to different development environments and languages.
 - Actionable feedback for developers to fix issues efficiently.
 - Integration capabilities with popular IDEs and frameworks.
- The solution aims to boost software quality, mitigate security risks, reduce manual code review efforts, and enhance overall development productivity.

45. High-Throughput Parallelized UDP Transfer with Priority Scheduling and Adaptive Redundancy

Background

In air-gapped environments, TCP cannot be used and static Forward Error Correction (FEC) approaches are inefficient. When transferring very large numbers of files of varying sizes, ensuring reliable, near-lossless delivery at high throughput (1000s of files of varying sizes from KB's to GB's) requires smarter scheduling and redundancy mechanisms.

Description

The proposed system should monitor a designated folder, schedule files for transfer with priority given to smaller files over larger files (solely based on file sizes), and transmit them in parallel over multiple UDP ports for maximum throughput. A redundancy mechanism must be added that goes beyond static FEC, ensuring lossless transfer even under variable loss conditions. The proposed architecture includes two machines connected back to back with one machine designated as Transmitter (from where files will be sent) and the other machine designated as Receiver (where files will be received) and the entire communication must happen over UDP and should be strictly one way communication.

Expected Solution

- A working prototype that monitors a folder, schedules files based on priority, and transmits them in parallel using UDP.
- Integrated redundancy that ensures better reliability than static FEC while maintaining high throughput.
- Performance evaluation on large workloads (e.g., 10k files with varying sizes over 2 hours).